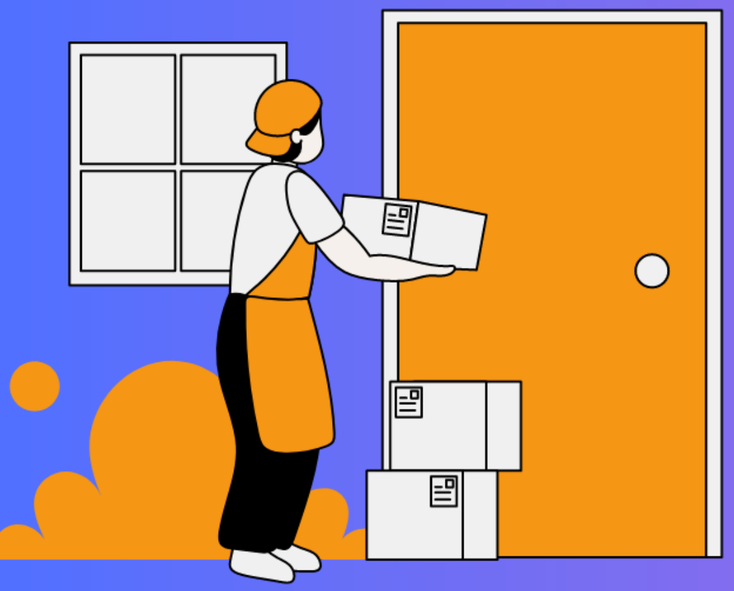
SQL Project Report

Online Food Delivery

Advance Level

What the project is about

- The project is focused on building an online food delivery system using sql database
- It aims to manage and organize data realated to restaurants menu items,customers,orders and its details
- Allows for efficient reporting and decision making through sturctured data and sql queries
- Calculate the total revenue generated by each restaurants and making its easy to identify top performing



ER Diagram

CUSTOMERS

customer_id int primary key
customer_name varchar(50),
email varchar(50),
city varchar(50),
signup_date date)

RESTAURANTS

restaurant_id int primary key,
restaurant_name varchar(50),
city varchar(50),
registration_date date)

MENU ITEMS

item_id int primary key,
restaurant_id (FK)
item_name varchar(50),
price float

ORDERS

order_id int primary key,
customer_id (FK),
restaurant_id (FK)
order_date date,

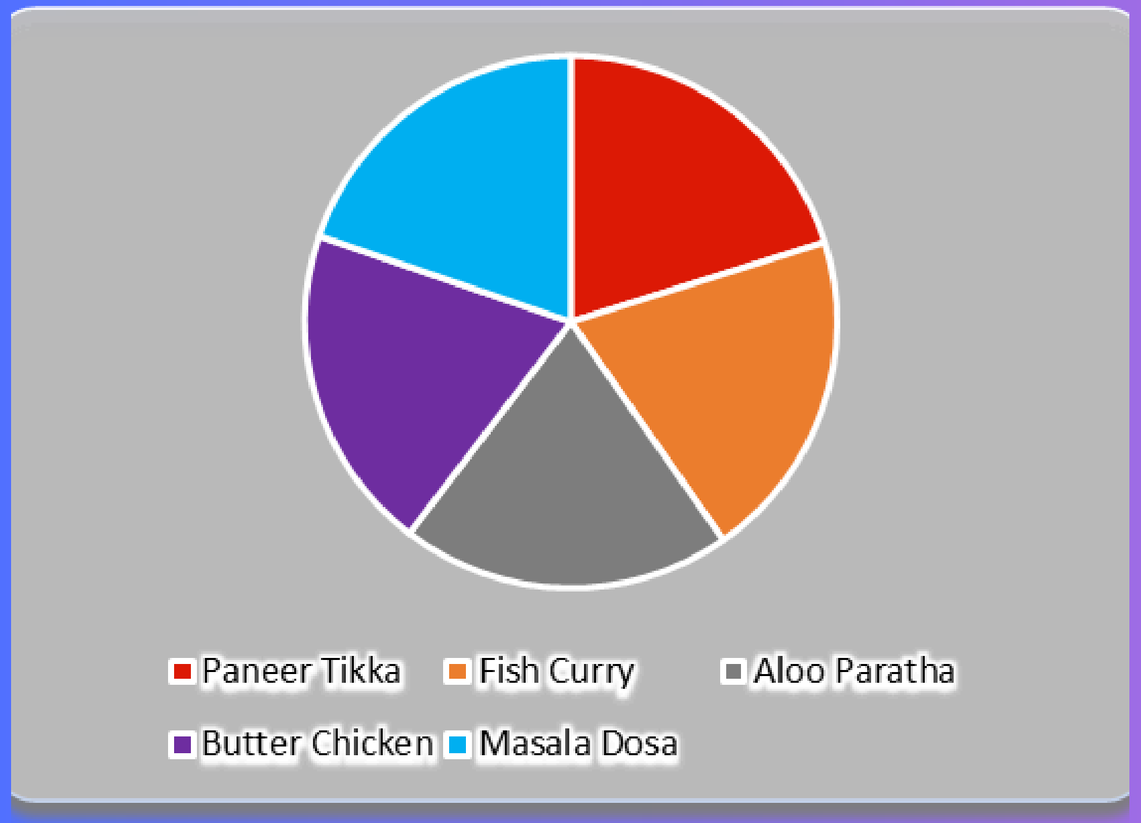
ORDER DETAILS

order_detail_id int primary key,
order_id (FK),
item_id int(FK)
quantity int,

Queries & Methods

Q1: Top 5 Most Expensive Menu Items

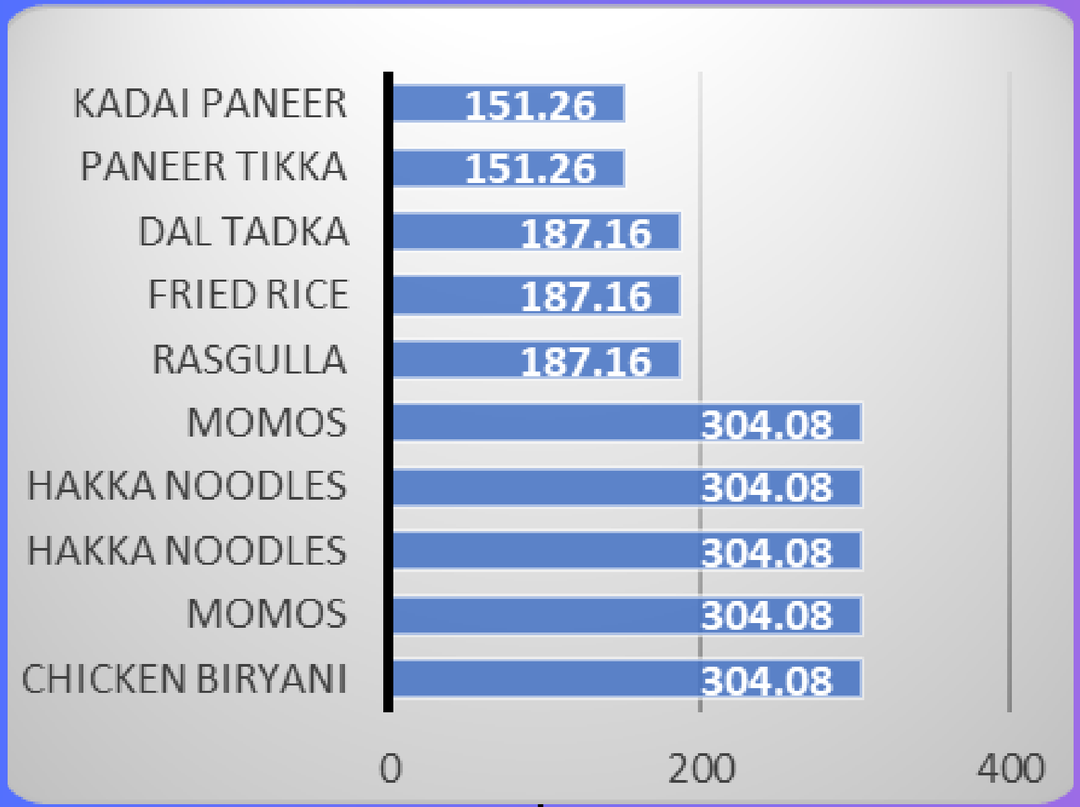
```
select item_id,item_name,sum(price) as total_price
from menu_items
group by item_id,item_name
order by total_price desc  limit 5;
```



item_id	item_name	total_price
79	Paneer Tikka	598.530029296875
8	Fish Curry	597.9000244140625
124	Aloo Paratha	596.9400024414062
134	Butter Chicken	588.8200073242188
22	Masala Dosa	587.0399780273438

Queries & Methods

Q2.Cheapest menu items by restaurant

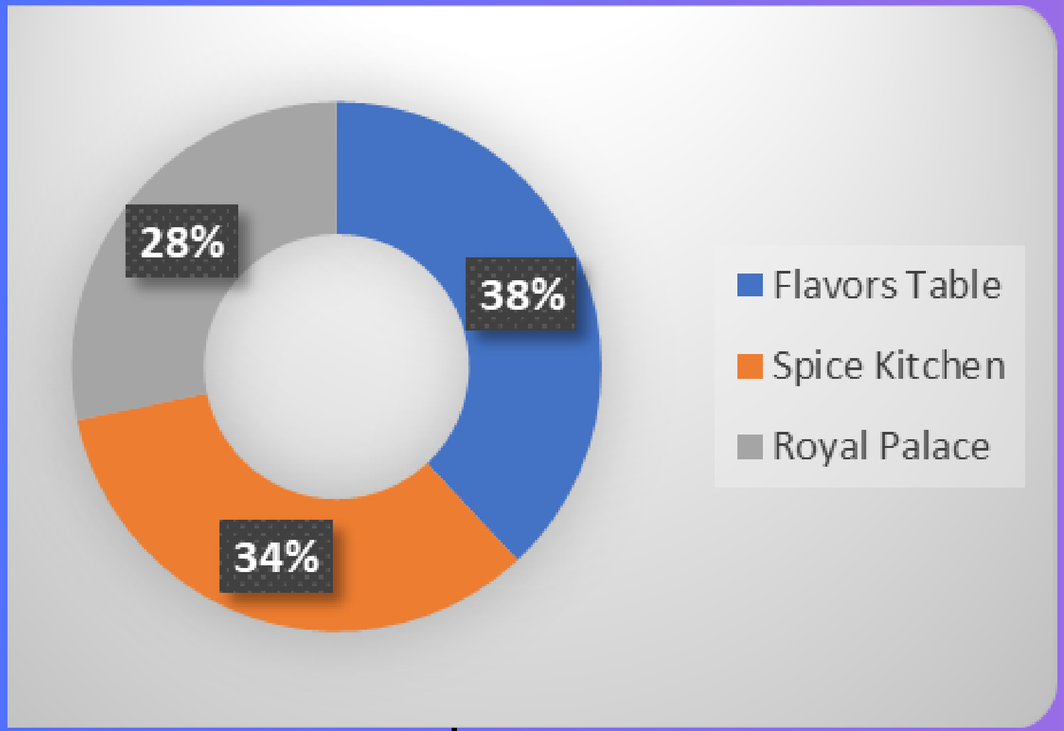



```
select restaurant_id,item_name,min(price) over
(partition by restaurant_id) as cheapest_item
from menu_items limit 10;
```

restaurant_id	item_name	cheapest_item
1	Chicken Biryani	304.08
1	Momos	304.08
1	Hakka Noodles	304.08
1	Hakka Noodles	304.08
1	Momos	304.08
2	Rasgulla	187.16
2	Fried Rice	187.16
2	Dal Tadka	187.16
3	Paneer Tikka	151.26
3	Kadai Paneer	151.26

Queries & Methods

Q3.Top 3 Revenue Generating Restaurants

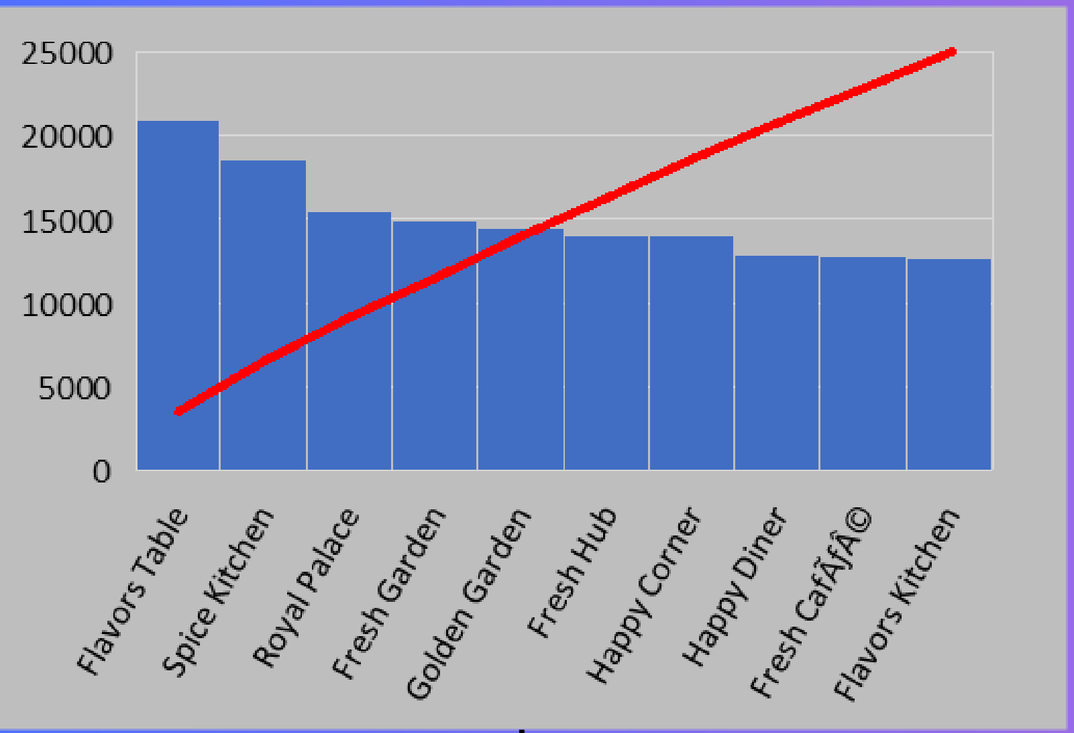



```
select r.restaurant_id,r.restaurant_name,
sum(od.quantity * m.price) as Total_revenue
 from restaurants r
join menu_items m on r.restaurant_id = m.restaurant_id
join order_details od on od.item_id = m.item_id
group by r.restaurant_id,r.restaurant_name
order by Total_revenue desc limit 3;
```

restaurant_id	restaurant_name	Total_revenue
30	Flavors Table	20906.109619140625
47	Spice Kitchen	18530.83009338379
16	Royal Palace	15528.30012512207

Queries & Methods

Q4.Restaurant Revenue Leaderboard

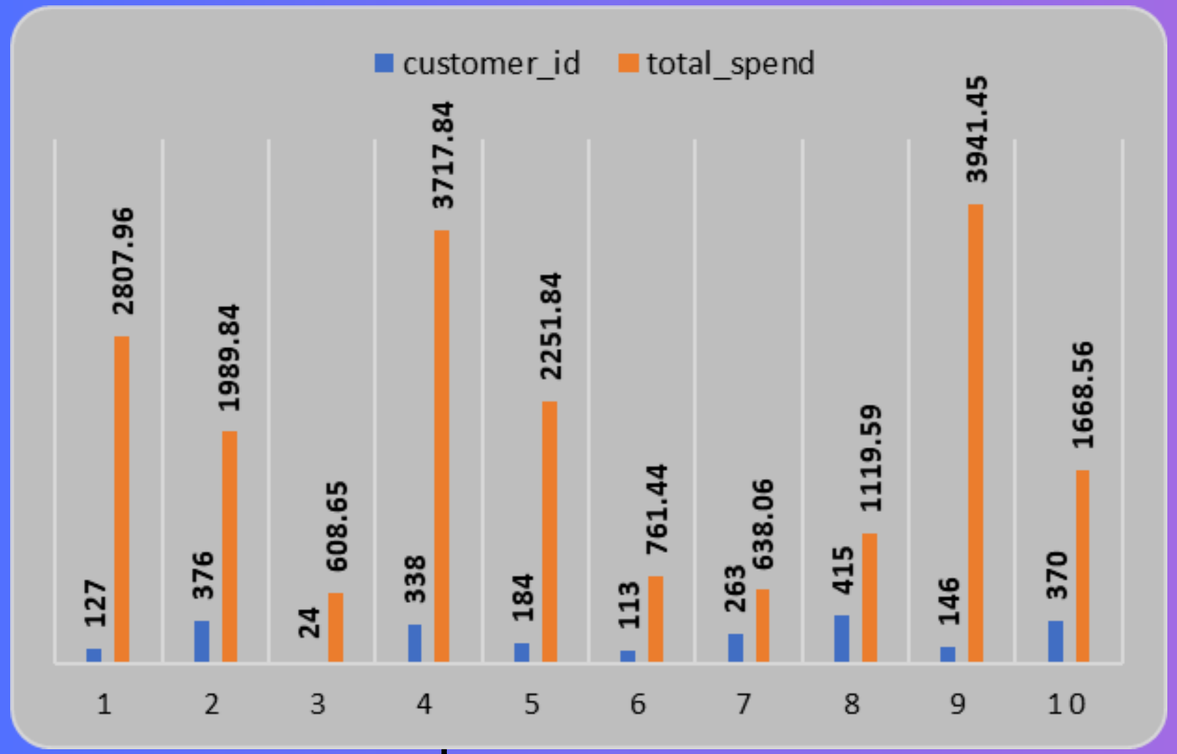



```
with restaurant_revenue_leaderboard as
(select r.restaurant_id,r.restaurant_name,sum(m.price*od.quantity) as
 total_revenue,rank() over
(partition by r.restaurant_id,r.restaurant_name
order by sum(m.price*od.quantity)desc) as rnk
from restaurants r
join menu_items m on r.restaurant_id = m.restaurant_id
join order_details od on m.item_id = od.item_id
group by r.restaurant_id,r.restaurant_name)
select * from restaurant_revenue_leaderboard
order by total_revenue desc limit 10;
```

restaurant_id	restaurant_name	total_revenue	rnk
30	Flavors Table	20906.109619140625	1
47	Spice Kitchen	18530.83009338379	1
16	Royal Palace	15528.30012512207	1
48	Fresh Garden	14912.0400390625	1
45	Golden Garden	14528.63996887207	1
1	Fresh Hub	14032.300018310547	1
23	Happy Corner	13988.860107421875	1
42	Happy Diner	12888.689971923828	1

Queries & Methods

Q5.Average Total Value Spend By Customer.



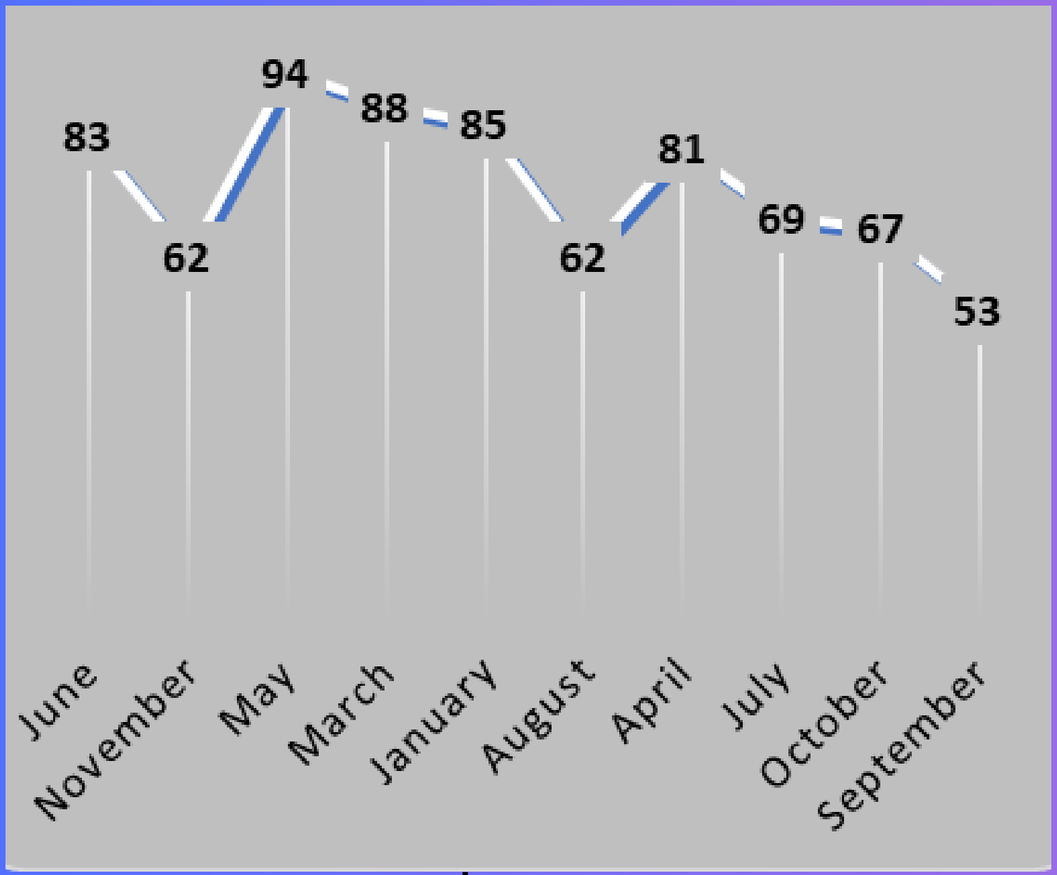

```
create view avg_spend_per_order as
select o.order_id,c.customer_id,sum(m.price * od.quantity)
as total_spend from orders o
join customers c on o.customer_id = c.customer_id
 join order_details od on o.order_id = od.order_id
join menu_items m on od.item_id = m.item_id
group by o.order_id,c.customer_id ;

select * from avg_spend_per_order limit 10;
```

order_id	customer_id	total_spend
379	127	2807.9600372314453
163	376	1989.8399658203125
275	24	608.650016784668
961	338	3717.8399505615234
377	184	2251.840087890625
1007	113	761.4400024414062
336	263	638.0599975585938
67	415	1119.5899963378906
553	146	3941.449920654297
456	370	1668.5599975585938

Queries & Methods

Q7.Monthly Order summary

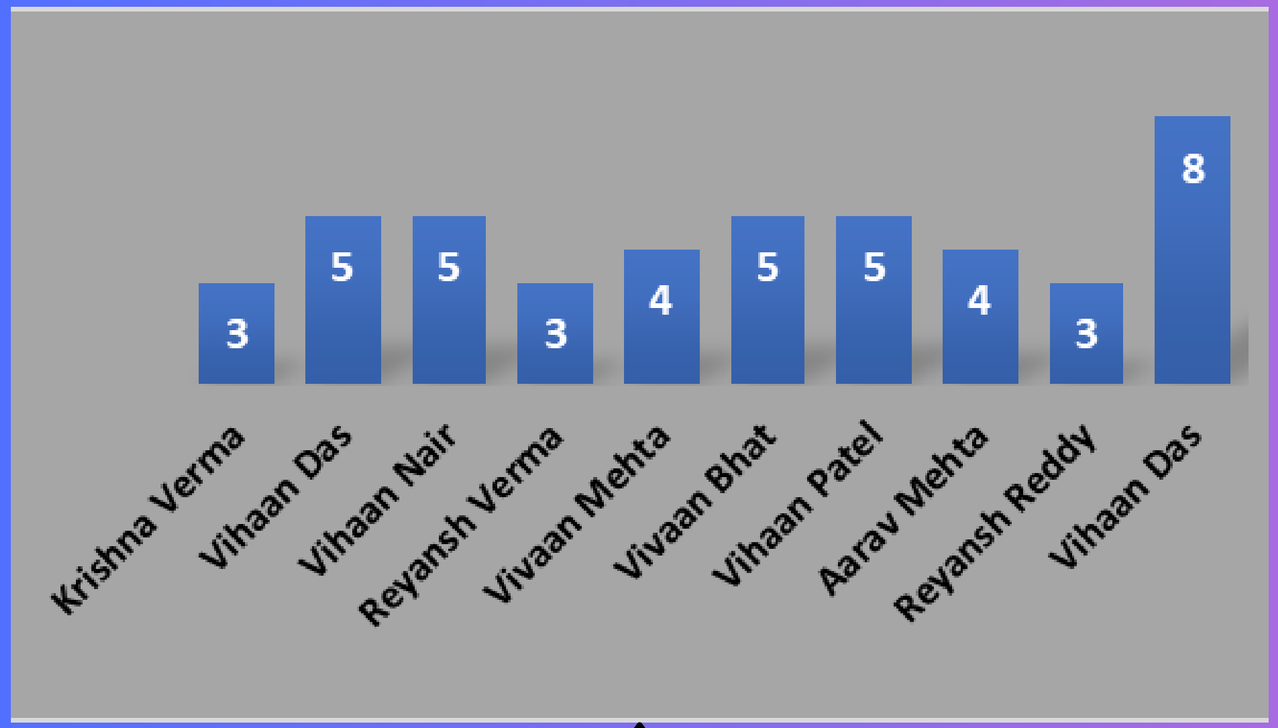



```
with monthly_order_summary as
(select extract(month from order_date)
as order_month,count(*) as total_orders
from orders
group by extract(month from order_date)
)
select order_month,total_orders
from monthly_order_summary
where total_orders > 50 limit 10;
```

order_month	total_orders
6	83
11	62
5	94
3	88
1	85
8	62
4	81
7	69
10	67
9	53

Queries & Methods

Q6.Customer With More Than 2 Orders

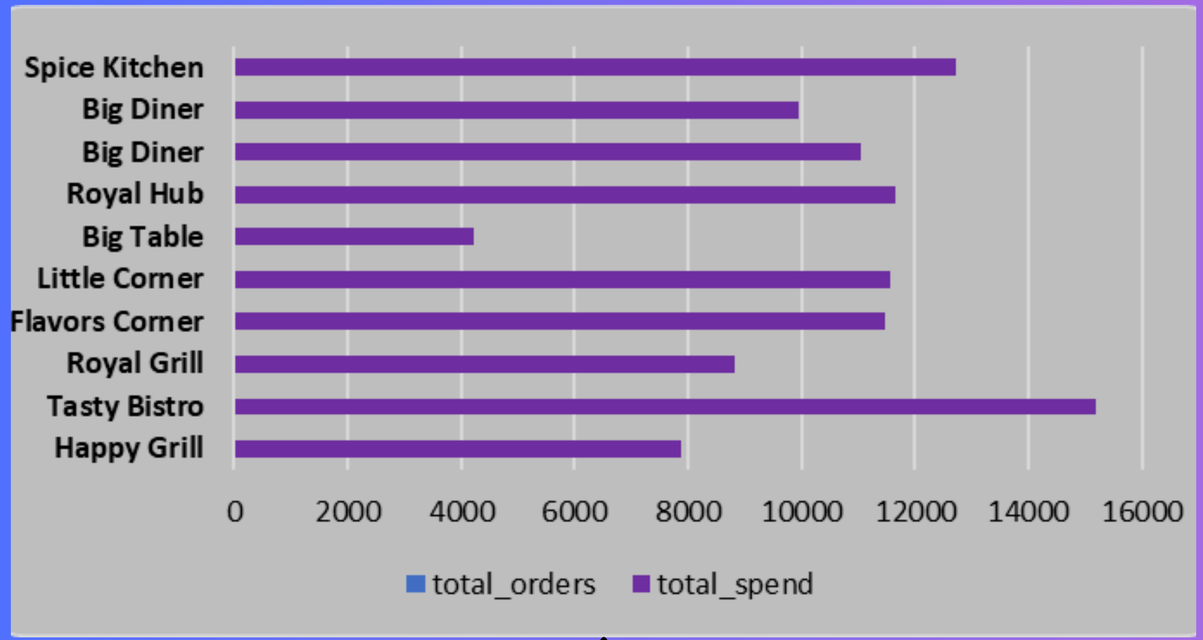



```sql
select customer_id,customer_name,order_count from (
select c.customer_id,c.customer_name,count(o.order_id)
as order_count from customers c
join orders o on  c.customer_id = o.customer_id
group by c.customer_id,c.customer_name ) as sub
having order_count >2 limit 10;
```

customer_id	customer_name	order_count
3	Krishna Verma	3
5	Vihaan Das	5
7	Vihaan Nair	5
10	Reyansh Verma	3
11	Vivaan Mehta	4
20	Vivaan Bhat	5
21	Vihaan Patel	5
24	Aarav Mehta	4
26	Reyansh Reddy	3
28	Vihaan Das	8

Queries & Methods

Q8.Restaurant Performance

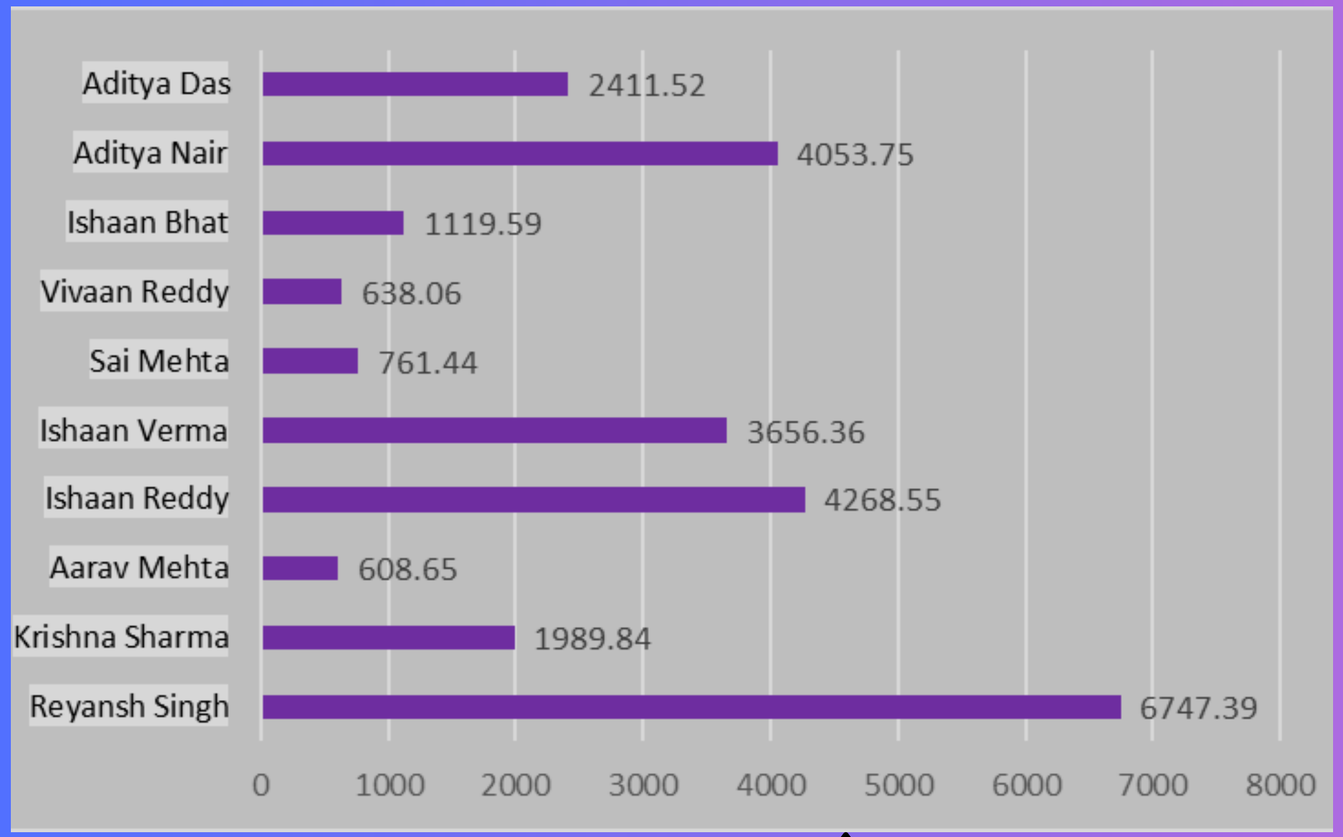



```sql
create view restaurant_performance as
select r.restaurant_id,r.restaurant_name,count(o.order_id) as total_orders,
sum(m.price * od.quantity) as total_spend
from restaurants r
join orders o on r.restaurant_id = o.restaurant_id
join order_details od on o.order_id = od.order_id
join menu_items m on od.item_id = m.item_id
group by r.restaurant_id,r.restaurant_name;
```

restaurant_id	restaurant_name	total_orders	total_spend
43	Happy Grill	7	7857.999984741211
6	Tasty Bistro	12	15168.52001953125
31	Royal Grill	6	8815.279899597168
10	Flavors Corner	11	11444.7099609375
50	Little Corner	10	11554.999954223633
27	Big Table	8	4210.089996337891
4	Royal Hub	9	11631.61996459961
46	Big Diner	16	11024.599967956543
2	Big Diner	8	9940.939910888672
47	Spice Kitchen	12	12704.159927368164

Queries & Methods

Q9.Total Spend By Customer



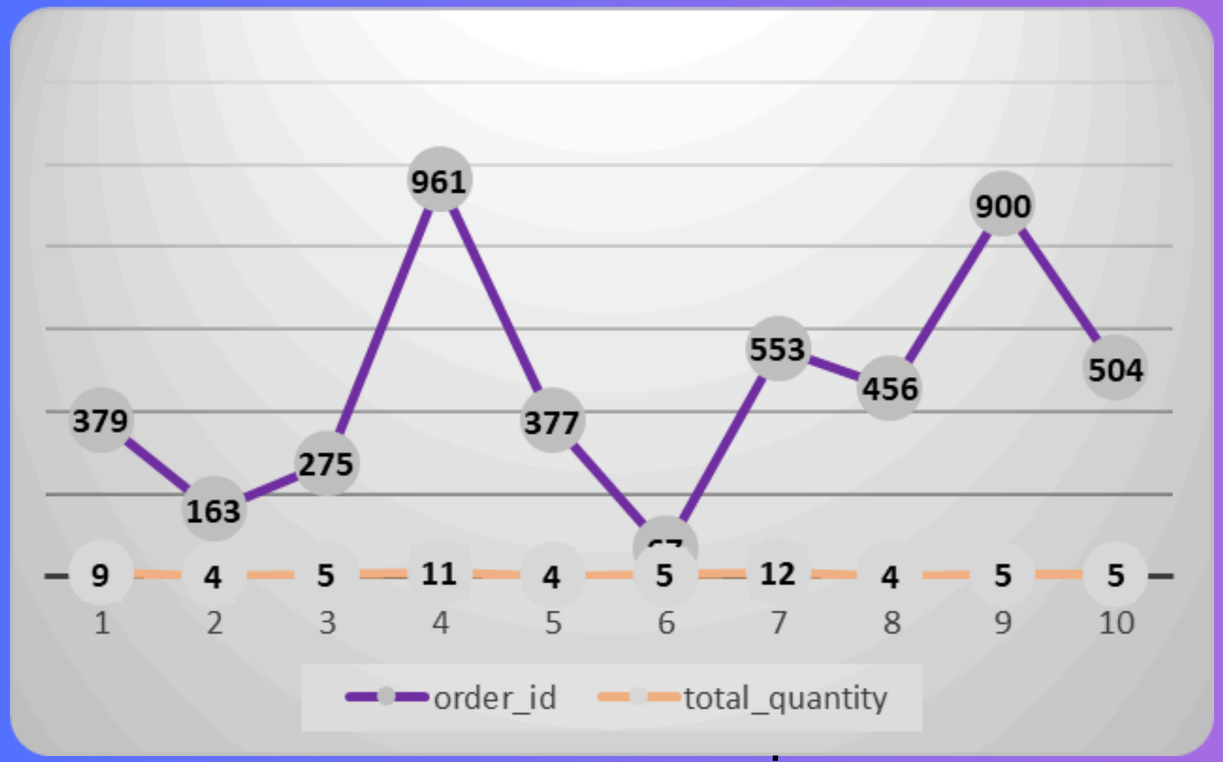

```sql
create temporary table temp_total_spending as
select c.customer_id,c.customer_name,sum(m.price*od.quantity)
 as total_spent from customers c
join orders o on c.customer_id = o.customer_id
join order_details od on o.order_id = od.order_id
join menu_items m on od.item_id = m.item_id
group by c.customer_id,c.customer_name;

select * from temp_total_spending
order by total_spent desc limit 10;
```

customer_id	customer_name	total_spent
489	Vihaan Singh	10824.400024414062
213	Reyansh Patel	9313.889923095703
138	Vihaan Patel	6768.299957275391
127	Reyansh Singh	6747.38996887207
453	Arjun Singh	6730.980072021484
56	Aarav Mehta	6521.340240478516
425	Sai Das	6379.18994140625
439	Sai Nair	6284.630020141602
408	Aditya Verma	5452.44010925293
234	Vihaan Gupta	5302.240051269531

Queries & Methods

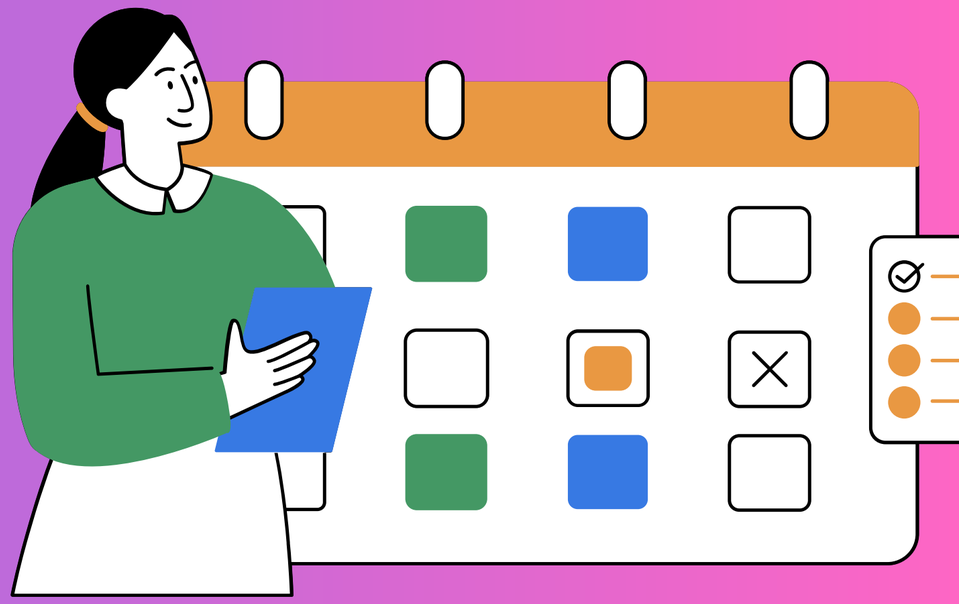
Q10.Big Order Quantity




```
with big_orders_qty as
(select o.order_id,sum(od.quantity) as total_quantity
from orders o join order_details od
on o.order_id = od.order_id group by o.order_id)
select * from big_orders_qty bq
where bq.total_quantity > 3 limit 10;
```

order_id	total_quantity
379	9
163	4
275	5
961	11
377	4
67	5
553	12
456	4
900	5
504	5

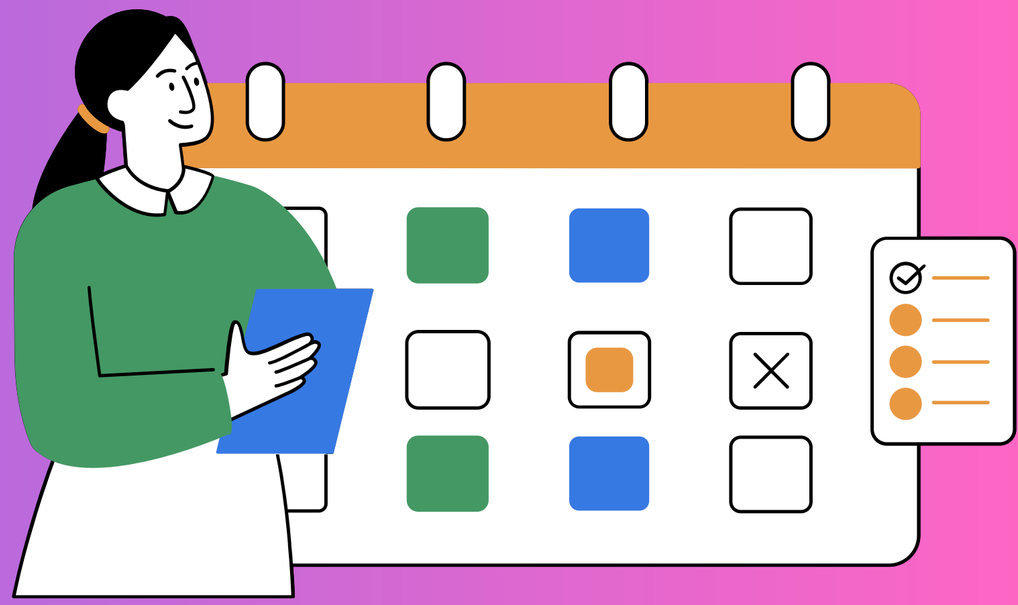
Insights & Storytelling



- Panner Tikka is the most expensive menu items
- Masala dosa and Butter Chicken also rank high in price
- Chicken Briyani and Noodles appear as the lowest priced items
- Most cheapest items fall in the ₹300 -₹350 range
- Flavors table is the top revenue generating restaurants

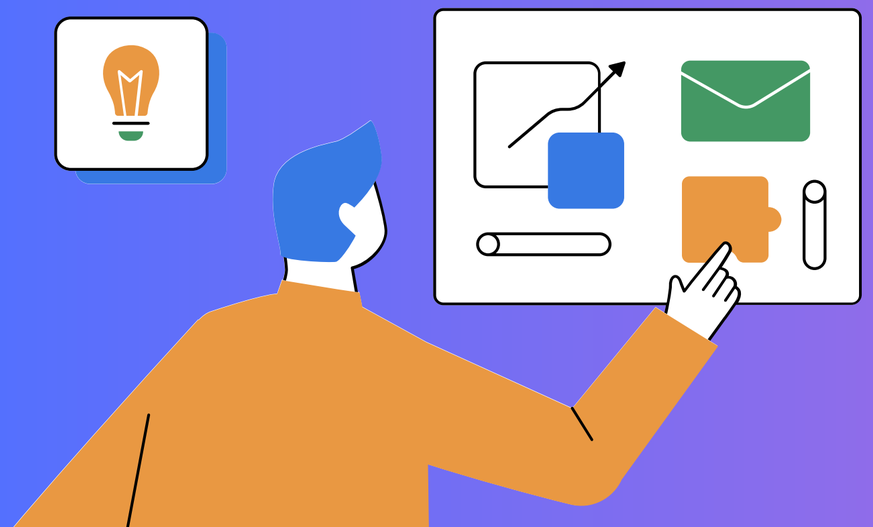
Insights & Storytelling

- May had the highest orderrs(94),september the lowest (53)
- Reyansh Singh spent the most, totaling 6747.39 making him the top customer by spending
- Krishna Verma and Vihan Das placed the most orders (9 each) ,showing high engagement
- the largest order had a total quantity of 9, while several other orders had a quantities between 5 and 8
- Most months had order counts above 60,with noticeable drop in November and september



Conclusion & Recommendations

- order volumes shows monthly fluctations, with peaks in may and lows in september. A loyal core of customers drives frequently and high value orders
- Focus marketing around may to capitalize on high order activity .
- Target less active months(like september and november) with promotions
- Engage top spenders and frequent buyers with loyalty programs to maintain and boast sales



Thank You

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